

Time Series — Solutions

Practice Exam — Week 13: Linear Time-Invariant (LTI) Filters

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Problem 1 (5 points)

Statement.

(LTI axioms, impulse response and transfer function.) A digital filter L maps an input sequence to an output sequence, $\{Y_t\} = L[\{X_t\}]$; we write a single output entry as $Y_u = L[\{X_t\}]_u$. Recall that L is linear time-invariant (LTI) if it is linear and commutes with the backshift, $LB = BL$. The impulse response is $h_m = L[\{\delta_{t,-m}\}]_0$ and, for $h \in \ell^1$, the transfer function is $H(f) = \sum_{m \in \mathbb{Z}} h_m e^{-2\pi i f m}$.

(a) Let L_1, L_2 be LTI filters and $\alpha \in \mathbb{C}$. Prove, directly from the two axioms, that the linear combination $L = \alpha L_1 + L_2$ is again LTI (check linearity and time invariance separately). (2 pts)

(b) Consider the concrete one-sided filter

$$L[\{X_t\}]_u = X_u - \frac{1}{2} X_{u-1}, \quad u \in \mathbb{Z}.$$

Identify its impulse response $\{h_m\}$, write L as a polynomial $\Phi(B)$ in the backshift, and compute its transfer function $H(f)$ together with the squared gain $|H(f)|^2$ in real form (no complex exponentials). (2 pts)

(c) At which frequency $f \in [0, \frac{1}{2}]$ is the squared gain $|H(f)|^2$ of part (b) largest, and at which is it smallest? Give both extreme values of $|H(f)|^2$, and state in one sentence whether this filter is low-pass or high-pass. (1 pt)

Solution.

(a) We verify the two axioms on a generic input, using only the linearity and time invariance of L_1, L_2 .

Linearity. For sequences $\{X_t\}, \{Y_t\}$ and $\beta \in \mathbb{C}$,

$$\begin{aligned} L[\beta\{X_t\} + \{Y_t\}] &= \alpha L_1[\beta\{X_t\} + \{Y_t\}] + L_2[\beta\{X_t\} + \{Y_t\}] \\ &= \alpha(\beta L_1[\{X_t\}] + L_1[\{Y_t\}]) + (\beta L_2[\{X_t\}] + L_2[\{Y_t\}]) \\ &= \beta(\alpha L_1[\{X_t\}] + L_2[\{X_t\}]) + (\alpha L_1[\{Y_t\}] + L_2[\{Y_t\}]) \\ &= \beta L[\{X_t\}] + L[\{Y_t\}]. \end{aligned}$$

Time invariance. Using that each L_j commutes with B and that B is linear,

$$\begin{aligned} L[B[\{X_t\}]] &= \alpha L_1[B[\{X_t\}]] + L_2[B[\{X_t\}]] = \alpha B[L_1[\{X_t\}]] + B[L_2[\{X_t\}]] \\ &= B[\alpha L_1[\{X_t\}] + L_2[\{X_t\}]] = B[L[\{X_t\}]]. \end{aligned}$$

Both axioms hold, so $L = \alpha L_1 + L_2$ is LTI. $\square$ (In particular the LTI filters form a vector space over $\mathbb{C}$, so every polynomial in B is LTI.)

(b) Writing $L[\{X_t\}]_u = \sum_k h_k X_{u-k}$, the coefficient of X_u is h_0 and the coefficient of X_{u-1} is h_1 . Comparing with $X_u - \frac{1}{2} X_{u-1}$,

$$h_0 = 1, \quad h_1 = -\frac{1}{2}, \quad h_m = 0 \quad (m \neq 0, 1).$$

Thus $L = I - \frac{1}{2}B = \Phi(B)$ with $\Phi(z) = 1 - \frac{1}{2}z$. Its transfer function is the Fourier transform of h (equivalently the polynomial-filter lemma $H = \Phi(e^{-2\pi i f})$):

$$H(f) = \sum_m h_m e^{-2\pi i f m} = 1 - \frac{1}{2} e^{-2\pi i f}.$$

The squared gain is

$$\begin{aligned} |H(f)|^2 &= \left(1 - \frac{1}{2}e^{-2\pi if}\right)\left(1 - \frac{1}{2}e^{2\pi if}\right) = 1 - \frac{1}{2}(e^{2\pi if} + e^{-2\pi if}) + \frac{1}{4} \\ &= \frac{5}{4} - \cos(2\pi f). \end{aligned}$$

(c) On $f \in [0, \frac{1}{2}]$, $\cos(2\pi f)$ decreases monotonically from 1 (at $f = 0$) to -1 (at $f = \frac{1}{2}$), so $|H(f)|^2 = \frac{5}{4} - \cos(2\pi f)$ is *smallest* at $f = 0$ and *largest* at $f = \frac{1}{2}$:

$$|H(0)|^2 = \frac{5}{4} - 1 = \frac{1}{4} \quad (\text{minimum}), \quad \left|H\left(\frac{1}{2}\right)\right|^2 = \frac{5}{4} + 1 = \frac{9}{4} \quad (\text{maximum}).$$

Since the gain is smallest at low frequency and largest at high frequency, the filter is **high-pass**: it attenuates the slowly varying (low-frequency) part of the series and emphasises rapid oscillations.

Problem 2 (5 points)

Statement.

(Differencing, smoothing, and the output spectrum.) Let $\{X_t\}$ be a zero-mean stationary process with $\gamma^{(X)} \in \ell^1$ and spectral density $\mathcal{S}_X(f)$. Recall the output-spectrum theorem: if $\{Y_t\} = L[\{X_t\}]$ for an LTI filter with transfer function H , then $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$.

(a) Let $\{Y_t\} = (I - B)[\{X_t\}]$, i.e. $Y_t = X_t - X_{t-1}$. Explain briefly why $\{Y_t\}$ is stationary, then prove that

$$\mathcal{S}_Y(f) = 4 \sin^2(\pi f) \mathcal{S}_X(f).$$

(Show the algebraic step $|1 - e^{-2\pi if}|^2 = 4 \sin^2(\pi f)$.) (2 pts)

(b) Now cascade the difference with the centred three-point average

$$L_2[\{V_t\}]_u = \frac{1}{3}(V_{u-1} + V_u + V_{u+1}),$$

forming $\{W_t\} = L_2[(I - B)[\{X_t\}]]$. Using the fact that the transfer function of a cascade is the product of the two transfer functions, express $\mathcal{S}_W(f)$ in terms of $\mathcal{S}_X(f)$, and name every frequency in $[0, \frac{1}{2}]$ at which $\mathcal{S}_W$ is forced to vanish. (2 pts)

(c) Specialise to white-noise input, $\mathcal{S}_X(f) \equiv \sigma^2$, and evaluate $\mathcal{S}_Y(\frac{1}{4})$ for the differenced series of part (a). (1 pt)

Solution.

(a) The filter $I - B$ has impulse response $h_0 = 1$, $h_1 = -1$ (all others 0), so $h \in \ell^1$. Filtering a stationary process by an ℓ^1 filter yields a stationary process (filtering preserves stationarity), hence $\{Y_t\}$ is stationary and has a spectral density. Its transfer function is

$$H(f) = h_0 + h_1 e^{-2\pi if} = 1 - e^{-2\pi if}.$$

By the output-spectrum theorem $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$, and

$$\begin{aligned} |1 - e^{-2\pi if}|^2 &= (1 - \cos 2\pi f)^2 + \sin^2 2\pi f = 1 - 2 \cos 2\pi f + \cos^2 2\pi f + \sin^2 2\pi f \\ &= 2 - 2 \cos(2\pi f) = 4 \sin^2(\pi f), \end{aligned}$$

using $\cos^2 + \sin^2 = 1$ and the half-angle identity $1 - \cos(2\theta) = 2 \sin^2 \theta$ with $\theta = \pi f$. Therefore

$$\mathcal{S}_Y(f) = 4 \sin^2(\pi f) \mathcal{S}_X(f).$$

Note $H(0) = 0$: differencing kills the $f = 0$ component, the signature of a high-pass operation that removes a constant level / linear trend.

(b) The three-point average has coefficients $h_{-1} = h_0 = h_1 = \frac{1}{3}$ (it is symmetric, zero phase), so its transfer function is real:

$$H_2(f) = \frac{1}{3}(e^{2\pi if} + 1 + e^{-2\pi if}) = \frac{1}{3}(1 + 2\cos 2\pi f).$$

The cascade $\{W_t\} = L_2[(I - B)[\{X_t\}]]$ is LTI with transfer function the *product*

$$H(f) = \underbrace{(1 - e^{-2\pi if})}_{\text{difference}} \cdot \underbrace{\frac{1}{3}(1 + 2\cos 2\pi f)}_{\text{3-point average}},$$

so the squared gains multiply:

$$\boxed{\mathcal{S}_W(f) = |H(f)|^2 \mathcal{S}_X(f) = 4\sin^2(\pi f) \cdot \frac{1}{9}(1 + 2\cos 2\pi f)^2 \mathcal{S}_X(f).}$$

The factor $4\sin^2(\pi f)$ vanishes at $f = 0$ (the difference). The factor $(1 + 2\cos 2\pi f)^2$ vanishes where $\cos(2\pi f) = -\frac{1}{2}$, i.e. $2\pi f = \frac{2\pi}{3}$, that is $f = \frac{1}{3}$. Hence on $[0, \frac{1}{2}]$ the output spectrum is forced to be zero at

$$f = 0 \quad \text{and} \quad f = \frac{1}{3}.$$

(The difference removes the DC component, the average annihilates $f = \frac{1}{3}$: a band-suppressing combination.)

(c) With $\mathcal{S}_X(f) \equiv \sigma^2$ and part (a),

$$\mathcal{S}_Y(\frac{1}{4}) = 4\sin^2(\frac{\pi}{4})\sigma^2 = 4 \cdot \left(\frac{1}{\sqrt{2}}\right)^2 \sigma^2 = 4 \cdot \frac{1}{2} \sigma^2 = 2\sigma^2.$$

Problem 3 (5 points)

Statement.

(ARMA spectral density and time reversibility of AR(p).) For a stationary ARMA(p, q) process $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the spectral density is

$$\mathcal{S}_X(f) = \sigma^2 \frac{|\Theta(e^{-2\pi if})|^2}{|\Phi(e^{-2\pi if})|^2}, \quad \Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j, \quad \Theta(z) = 1 - \sum_{j=1}^q \theta_j z^j.$$

(a) For the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, use the polynomial-filter lemma (the filter $\Phi(B)$ has transfer function $\Phi(e^{-2\pi if})$) to derive

$$\mathcal{S}_X(f) = \frac{\sigma^2}{1 + \phi^2 - 2\phi \cos(2\pi f)},$$

and state, with reason, where $\mathcal{S}_X$ peaks when $\phi > 0$ and when $\phi < 0$. (2 pts)

(b) Let $\{X_t\}$ be a Gaussian AR(p) process, $X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t$, and define the time-reversed series $Y_t = X_{-t}$. Show that $\gamma_\tau^{(Y)} = \gamma_\tau^{(X)}$ for all τ , hence $\mathcal{S}_Y(f) = \mathcal{S}_X(f)$. (1.5 pts)

(c) Put $\phi_0 = -1$ and $\tilde{v}_t = \sum_{j=0}^p \phi_j Y_{t-j} = \Phi(B)[\{Y_t\}]_t$. Using parts (a)–(b), the polynomial-filter lemma and the output-spectrum theorem, prove that $\mathcal{S}_{\tilde{v}}(f) \equiv \sigma^2$, and conclude that $\{Y_t\}$ is an AR(p) with the same coefficients $\phi_1, \dots, \phi_p$ driven by white noise of variance σ^2 . (1.5 pts)

Solution.

(a) Rewrite the AR(1) as $\Phi(B)X_t = \varepsilon_t$ with $\Phi(z) = 1 - \phi z$. By the polynomial-filter lemma the filter $\Phi(B)$ has transfer function $\Phi(e^{-2\pi if}) = 1 - \phi e^{-2\pi if}$, so applying the output-spectrum theorem to $\Phi(B)[\{X_t\}] = \{\varepsilon_t\}$ (with $\mathcal{S}_\varepsilon \equiv \sigma^2$),

$$\left|1 - \phi e^{-2\pi if}\right|^2 \mathcal{S}_X(f) = \sigma^2.$$

Expanding the modulus,

$$\left|1 - \phi e^{-2\pi i f}\right|^2 = (1 - \phi e^{-2\pi i f})(1 - \phi e^{2\pi i f}) = 1 + \phi^2 - \phi(e^{2\pi i f} + e^{-2\pi i f}) = 1 + \phi^2 - 2\phi \cos(2\pi f),$$

which is $\geq (1 - |\phi|)^2 > 0$ since $|\phi| < 1$ (so we may divide), giving

$$\boxed{\mathcal{S}_X(f) = \frac{\sigma^2}{1 + \phi^2 - 2\phi \cos(2\pi f)}}.$$

The denominator is smallest where $\phi \cos(2\pi f)$ is largest. For $\phi > 0$ that is $f = 0$, so $\mathcal{S}_X$ **peaks at** $f = 0$ (a slowly varying, persistent series); for $\phi < 0$ it is $f = \frac{1}{2}$, so $\mathcal{S}_X$ **peaks at** $f = \frac{1}{2}$ (a rapidly oscillating series).

(b) Since $\{X_t\}$ is stationary its ACVS is even, $\gamma_{-\tau}^{(X)} = \gamma_{\tau}^{(X)}$. With $Y_t = X_{-t}$ and using stationarity ($\mathbb{E}[Y_t] = 0$),

$$\gamma_{\tau}^{(Y)} = \text{Cov}(Y_{t+\tau}, Y_t) = \mathbb{E}[Y_{t+\tau}Y_t] = \mathbb{E}[X_{-t-\tau}X_{-t}] = \gamma_{(-t)-(-t-\tau)}^{(X)} = \gamma_{\tau}^{(X)},$$

where the lag is (first index) – (second index) = τ . Hence $\gamma_{\tau}^{(Y)} = \gamma_{\tau}^{(X)}$ for all τ , and taking Fourier transforms,

$$\mathcal{S}_Y(f) = \sum_{\tau} \gamma_{\tau}^{(Y)} e^{-2\pi i f \tau} = \sum_{\tau} \gamma_{\tau}^{(X)} e^{-2\pi i f \tau} = \mathcal{S}_X(f), \quad \forall f.$$

(c) The sequence $\{\tilde{v}_t\} = \Phi(B)[\{Y_t\}]$ is an LTI filtering of $\{Y_t\}$ with the polynomial filter $\Phi(B)$, $\Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j$. By the polynomial-filter lemma and the output-spectrum theorem,

$$\mathcal{S}_{\tilde{v}}(f) = \left| \Phi(e^{-2\pi i f}) \right|^2 \mathcal{S}_Y(f) \stackrel{(b)}{=} \left| \Phi(e^{-2\pi i f}) \right|^2 \mathcal{S}_X(f).$$

But $\{X_t\}$ is $\text{AR}(p)$, i.e. the $q = 0$ ARMA case, so $\mathcal{S}_X(f) = \sigma^2 / |\Phi(e^{-2\pi i f})|^2$ (the multivariate analogue of part (a); the denominator is nonzero on the unit circle by stationarity). Substituting,

$$\mathcal{S}_{\tilde{v}}(f) = \left| \Phi(e^{-2\pi i f}) \right|^2 \cdot \frac{\sigma^2}{|\Phi(e^{-2\pi i f})|^2} = \sigma^2, \quad \forall f.$$

A constant spectral density means $\{\tilde{v}_t\}$ is white noise of variance σ^2 ; being a linear combination of jointly Gaussian variables it is Gaussian with mean 0 (as $\mathbb{E}[X_t] = 0$), so it has the same law as $\{\varepsilon_t\}$. Finally, from $\phi_0 = -1$,

$$\tilde{v}_t = \sum_{j=0}^p \phi_j Y_{t-j} = -Y_t + \sum_{j=1}^p \phi_j Y_{t-j} \iff Y_t = \sum_{j=1}^p \phi_j Y_{t-j} - \tilde{v}_t.$$

Since $-\tilde{v}_t$ is again white noise of variance σ^2 with the same distribution as ε_t , the reversed series $\{Y_t\}$ is an $\text{AR}(p)$ with the *same* coefficients $\phi_1, \dots, \phi_p$. $\square$

Problem 4 (5 points)

Statement.

(Revision of Week 12: ARCH(1) — moments, white noise, and the ACF of the squares.)

Let $\{r_t\}$ be an ARCH(1) process,

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2,$$

with $\alpha_0 > 0$, $0 \leq \alpha_1 < 1$, and $\{\varepsilon_t\}$ i.i.d. standard Gaussian (mean 0, variance 1) independent of $\mathcal{F}_{t-1}$.

(a) Using that σ_t is $\mathcal{F}_{t-1}$ -measurable and iterated expectation, show $\mathbb{E}[r_t] = 0$, and (assuming weak stationarity) that

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

(1.5 pts)

(b) Prove that $\{r_t\}$ is serially uncorrelated, i.e. $\text{Cov}(r_{t+\tau}, r_t) = 0$ for all $\tau \neq 0$, by conditioning on $\mathcal{F}_{t+\tau-1}$. (This is what makes ARCH a white noise.)

(1.5 pts)

(c) Assume in addition $0 < \alpha_1 < 1/\sqrt{3}$ (so a finite fourth moment exists). Recall $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ and the kurtosis $\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2}$. Take $\alpha_1 = \frac{1}{2}$ and report the numerical value of $\text{Corr}(r_{t+1}^2, r_t^2)$, of $\text{Corr}(r_{t+2}^2, r_t^2)$, and of the kurtosis κ ; state in one sentence why $\kappa > 3$ means fat tails.

(2 pts)

Solution.

(a) Since σ_t is a function of the past (it depends on r_{t-1}^2), it is $\mathcal{F}_{t-1}$ -measurable, while ε_t is independent of $\mathcal{F}_{t-1}$ with $\mathbb{E}[\varepsilon_t] = 0$. Hence

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t \varepsilon_t \mid \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0 \implies \mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] = 0.$$

For the variance, $\mathbb{E}[\varepsilon_t^2] = 1$ gives $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[\varepsilon_t^2] = \sigma_t^2$, so by iterated expectation, using $\mathbb{E}[r_t] = 0$ (whence $\mathbb{E}[r_t^2] = \text{Var}(r_t)$),

$$\text{Var}(r_t) = \mathbb{E}[r_t^2] = \mathbb{E}[\sigma_t^2] = \mathbb{E}[\alpha_0 + \alpha_1 r_{t-1}^2] = \alpha_0 + \alpha_1 \mathbb{E}[r_{t-1}^2] = \alpha_0 + \alpha_1 \text{Var}(r_{t-1}).$$

By weak stationarity $\text{Var}(r_t) = \text{Var}(r_{t-1}) =: v$, so $v = \alpha_0 + \alpha_1 v$, i.e. $(1 - \alpha_1)v = \alpha_0$, hence

$$\boxed{\text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1} \quad (\alpha_1 < 1).}$$

(b) Fix $\tau > 0$. As $\mathbb{E}[r_t] = 0$, $\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t]$. Both r_t and $\sigma_{t+\tau}$ are $\mathcal{F}_{t+\tau-1}$ -measurable (r_t is observed by time $t + \tau - 1$, and $\sigma_{t+\tau}$ depends on $r_{t+\tau-1}^2$), while $\varepsilon_{t+\tau}$ is independent of $\mathcal{F}_{t+\tau-1}$ with mean 0. Conditioning on $\mathcal{F}_{t+\tau-1}$ and pulling out the measurable factors,

$$\mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}[\mathbb{E}[\sigma_{t+\tau} \varepsilon_{t+\tau} r_t \mid \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0.$$

By symmetry of the covariance the same holds for $\tau < 0$. Therefore $\text{Cov}(r_{t+\tau}, r_t) = 0$ and $\text{Corr}(r_{t+\tau}, r_t) = 0$ for all $\tau \neq 0$: $\{r_t\}$ is white noise. $\square$

(c) With $\alpha_1 = \frac{1}{2}$ we have $\alpha_1 = 0.5 < 1/\sqrt{3} \approx 0.577$, so a finite fourth moment exists. The ACF of the squares is $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$, so

$$\text{Corr}(r_{t+1}^2, r_t^2) = \alpha_1 = \frac{1}{2} = 0.5, \quad \text{Corr}(r_{t+2}^2, r_t^2) = \alpha_1^2 = \frac{1}{4} = 0.25.$$

The kurtosis is

$$\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} = \frac{3(1 - \frac{1}{4})}{1 - 3 \cdot \frac{1}{4}} = \frac{3 \cdot \frac{3}{4}}{\frac{1}{4}} = \frac{9/4}{1/4} = 9.$$

A normal distribution has kurtosis exactly 3; $\kappa = 9 > 3$ means the standardised distribution of r_t has *heavier tails than the Gaussian* (excess kurtosis 6), so large $|r_t|$ occurs far more often than under normality — the fat-tailed behaviour observed in financial returns, generated here endogenously even though ε_t is Gaussian.